Bayesian MCMC Validation of the Structural Self-Energy Expansion Framework (SSEE-V3.6):

Model Comparison against ΛCDM and CPL using DESI DR2, Planck 2018, and Galaxy-Cluster Mass Data

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Abstract

We present a Bayesian validation of the dynamical sector of the Structural Self-Energy Expansion (SSEE-V3.6) framework [Almeida Vallejo, 2026] against DESI DR2 baryon acoustic oscillation data [Abdul-Karim et al., 2025], a compressed Planck 2018 prior [Planck Collaboration, 2020], and galaxy-cluster dynamical masses derived under a variable IGIMF baryonic baseline [Zhang et al., 2026]. Using geometric constants fixed algebraically by π and the golden ratio φ — with H_0 as the sole free parameter — we compare the model against ΛCDM and a free CPL parameterisation via posterior constraints, information criteria, and predictive checks, using EMCEE [Foreman-Mackey et al., 2013, Goodman & Weare, 2010] with $N_{\text{walkers}} = 100$, $N_{\text{steps}} = 25{,}000$ (2.5×10^6 total evaluations), and a same-bin correlated block-diagonal covariance approximation for the DESI DR2 BAO measurements; convergence diagnostics (including $N_{\text{eff}} = 4{,}402$ effective samples for SSEE) are reported in Table 5. The SSEE prediction $(w_0, w_a) = (-0.840, -0.670)$, fixed algebraically by the closed dictionary [Almeida Vallejo, 2026b] and not fitted to the data, lies within 0.24σ of the official DESI DR2 w_0w_aCDM posterior (DESI+CMB+Pantheon+; arXiv:2503.14738, eq. 26) in the w_0–w_a plane, and within 1.2–1.5σ of the DESY5/Union3/no-SN combinations (a parameter-free postdiction). The H_0 posterior is $67.95^{+0.40}_{-0.40}$ km s^{-1} Mpc^{-1}, obtained with the ω_m-direct CMB anchor as the H_0 prior ($H_0^{\text{alg}} = 67.962$; Paper 3, §3.2). The cluster-mass sector achieves $\chi^2_r = 0.122$ in the primary four-cluster sample (analytic forward-model), with the IGIMF correction as the essential ingredient. With (w_0, w_a) fixed algebraically and one free parameter H_0, SSEE is favoured by ΔBIC = 5.68 over ΛCDM and 5.59 over CPL, and predicts the DESI DR2 test set better in leave-one-out cross-validation — so the w_0–w_a prediction is genuinely preferred by current BAO data. The background geometry uses the *total* gravitating matter density $\Omega_m = \omega_m/h^2 = 0.30889$ (ω_m-direct; no matter factor, OP-8 dissolved), for which

the SSEE sound horizon matches ΛCDM ($r_d^{\rm SSEE}/r_d^{\Lambda\rm CDM} = 1.002$) and Ω_m agrees with Planck at 0.88σ. The cold sector $1 + w_0 = 0.160$ enters the perturbations (Papers 3 and 6) and the EFT coupling, *not* the background geometry; an earlier "21.3σ" figure came from comparing that partial sector against the Planck *total* Ω_m — a category error corrected here. The two-sector ϕ-DM framework of Papers 3–6 remains the physical account of how the total density partitions across scales.

Note on the background matter density: Throughout, the background geometry ($E(z)$, the sound horizon r_d, and all BAO distances) uses the *total* gravitating matter density $\Omega_m = \omega_m/h^2 = 0.30889$, built piece-by-piece from SSEE algebra (ω_m-direct; no matter factor, OP-8 dissolved; Paper 3). The bookkeeping value $1 + w_0 = 0.160$ is the dark-energy equation of state, *not* a matter density: it enters only the perturbations (the two-sector ϕ-DM construction of Papers 3 and 6) and the EFT coupling α_K, never $E(z)$. With the total density the SSEE sound horizon matches ΛCDM and model selection yields a single, unambiguous figure: ΔBIC $= 5.68$ favouring SSEE over ΛCDM and 5.59 over CPL (Section 8).

Contents

1 Introduction

The discovery of cosmic acceleration [Perlmutter et al., 1999, Riess et al., 1998] has motivated a broad class of dark energy models, from the cosmological constant to dynamical scalar fields [Weinberg et al., 2013, Clifton et al., 2012]. Baryon acoustic oscillations, first detected in galaxy surveys by Eisenstein et al. [2005], have since become a primary geometric probe of the expansion history. The SSEE framework [Almeida Vallejo, 2026] is a **minimal-parameter phenomenological framework** that encodes the cosmological background as algebraic combinations of π and $\varphi = (1+\sqrt{5})/2$. The background sector $(w_0, w_a, \Omega_{\rm DE}, \Omega_{m,\rm dyn})$ tested in this paper carries zero dimensionless fitted parameters, and no WIMP, QCD-axion, or SUSY dark-matter particle is introduced; phenomenological extensions in the dark-matter sector (Paper 6) introduce ansätze tracked as open problems (see Paper 1, §1.3, and `OPEN_PROBLEMS.md`). Paper 1, Appendix A provides a complete Lagrangian (k-essence + Israel–Stewart viscosity) derived algebraically from $\{\varphi, \pi\}$, with zero free parameters at the EFT background level. Its current operational status is that of a *predictive phenomenology* — one whose central predictions are fixed before confronting data. Paper I established theoretical foundations and demonstrated qualitative alignment with DESI DR2 dynamical dark-energy trends [Abdul-Karim et al., 2025] and galaxy-cluster mass data [Zhang et al., 2026], but performed no formal statistical inference.

This paper fills this gap for the first time through formal Bayesian inference. Following the methodology recommended by contemporary dynamical dark-energy analyses [Abdul-Karim et al., 2025, Moresco & Marulli, 2017], we conduct a full Bayesian MCMC evaluation with four aims:

1. Quantify whether SSEE remains competitive under formal Bayesian inference against ΛCDM and free CPL alternatives.
2. Identify which physical ingredients reproduce cluster masses.
3. Characterise the $\Omega_{\rm DE}^{\rm SSEE} \approx 0.840$ vs $\Omega_\Lambda \approx 0.685$ background tension and localise its origin.
4. Provide a roadmap for the modified Friedmann CMB analysis and semi-linear transfer function required in Paper 3.

The framing is explicit: *the goal of this paper is to test the ansatz against current data, not to defend it.* Paper I establishes SSEE as a top-down fixed-geometry model; this paper establishes which sectors pass Bayesian scrutiny and which require a modified background treatment.

2 SSEE Model Summary

2.1 Algebraic constants

From π and $\varphi = (1+\sqrt{5})/2$, the framework defines the structural viscosity KAL_0 and the CPL [Chevallier & Polarski, 2001, Linder, 2003] parameters as:

$$\mathrm{KAL}_0 = \frac{\pi+\varphi}{2} + \pi \approx 5.5214, \tag{1}$$

$$w_0^{\rm SSEE} = -\frac{T_r}{M_v} \approx -0.840, \quad w_a^{\rm SSEE} = -\frac{P_{sc}}{K_v} \approx -0.670. \tag{2}$$

All three quantities are fixed by geometry; no fitting is performed. Note that $w_0 + w_a = -1.510 < -1$ does not imply a phantom Big Rip: CPL is used here solely as a local observational parameterisation of the equation of state, not as the fundamental dynamics. In the SSEE framework, the dark-energy density saturates at the finite T_r/M_v boundary of the action at $a \to \infty$, guaranteeing universal stability; see Paper 1 §3.4 [Almeida Vallejo, 2026] for the full stability argument. The dark-energy density fraction is similarly fixed: $\Omega_{\rm DE}^{\rm SSEE} = T_r/M_v \approx 0.840$, leaving an effective matter fraction fixed algebraically:

$$\Omega_{m,\rm eff} = 1 - \Omega_{\rm DE}^{\rm SSEE} \approx 0.160. \tag{3}$$

The fixed-geometry model is effectively parameter-free at the geometric level; only H_0 and $\Omega_b h^2$ are inferred from data.

2.2 Modified Friedmann background

$$H^2(z) = H_0^2 \left[\Omega_{m,\rm eff}(1+z)^3 + \Omega_{\rm DE}^{\rm SSEE}\, f_{\rm DE}(z) \right], \tag{4}$$

where $f_{\rm DE}(z) = (1+z)^{3(1+w_0+w_a)} \exp[-3w_a z/(1+z)]$ is the standard CPL dark-energy evolution, here used solely to translate the fixed SSEE geometric parameters into observable distances. The only sampled cosmological parameter is H_0; all geometric quantities $(w_0, w_a, \Omega_{\rm DE}^{\rm SSEE}, \Omega_{m,\rm eff})$ are fixed algebraically as described in Section 2.

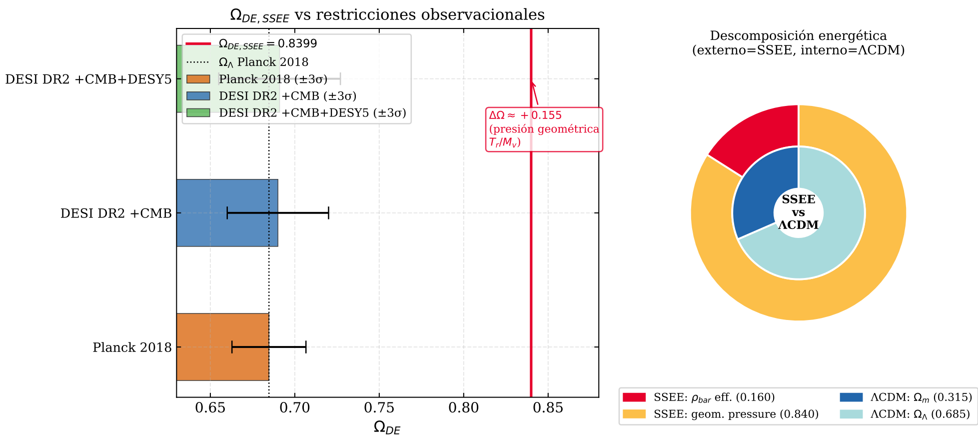

Figure 1: *Left*: $\Omega_{\rm DE}^{\rm SSEE} = T_r/M_v \approx 0.840$ (red line) compared with Planck 2018 $\Omega_\Lambda \approx 0.685$ (blue) and DESI DR2 $1\sigma/2\sigma$ bands. *Right*: Structural context showing the algebraic origin of $\Omega_{\rm DE}$ from the π–φ invariants.

2.3 Cluster mass formula

$$M_{\rm SSEE}(R) = M_{\rm bar}^{\rm IGIMF}(R) \cdot {\rm KAL}_0 \cdot (1 + f_\nu), \quad f_\nu \approx 0.020. \tag{5}$$

2.4 The sound horizon under the ω_m-direct reframe

Placing the cold sector alone $(1 + w_0 = 0.160)$ in the background geometry would give a drag scale $r_{d,\rm SSEE} \approx 175.6\,$Mpc, far from the Planck-inferred $r_d = 147.09 \pm 0.26\,$Mpc [Planck Collaboration, 2020]. That figure is a category error: $0.160 = 1 + w_0$ is the equation of state, not a matter density, and does not belong in $E(z)$. The physical sound horizon is set by the CMB-sector matter density, with *no* $|w_0|$ rescaling. The recombination matter budget is the standard physical $\omega_m = \omega_b + \omega_c + \omega_\nu = 0.14267$, built piece-by-piece from SSEE algebra ($\omega_c = \mathrm{KAL}_0\,\omega_b\,n_s$, forward; Paper 3), so that $\Omega_{m,\mathrm{CMB}} = \omega_m/h^2 = 0.30889$ and a CAMB integration of the recombination physics gives

$$r_d^{\omega_m\text{-direct}} = 147.17\ \mathrm{Mpc} \quad (0.32\sigma \text{ from Planck; Paper 3}), \tag{6}$$

with *no* matter factor. The earlier post-hoc mapping $r_{d,\mathrm{SSEE}} \times |w_0| \approx 147.5\,\mathrm{Mpc}$, and the structural ratio $\Omega_{m,\mathrm{CMB}}/\Omega_{m,\mathrm{dyn}} \approx \mathrm{MIRA} \approx 1.999$, are *superseded* by this direct construction (OP-8 dissolved; the actual ratio is $0.30889/0.160 = 1.93$, not a fixed factor). MIRA survives only as the Hubble screening amplitude of Paper 9. The two matter densities are now two *independent* algebraic predictions separated by the ϕ-DM free-streaming scale $k_{\mathrm{fs}} = 0.754\,h\,\mathrm{Mpc}^{-1}$ (Paper 6), not a factor pair. The falsifiable content therefore moves from the $|w_0|$ coincidence to the direct CMB fit of Paper 3 (canonical $\Delta\mathrm{BIC} = -32.9$ full `plik` MCMC; -23.9 `plik_lite` TTTEEE point estimate): if that fit had yielded $\chi^2_r \gg 1$ or excluded $r_d^{\omega_m\text{-direct}}$, the construction would be falsified.

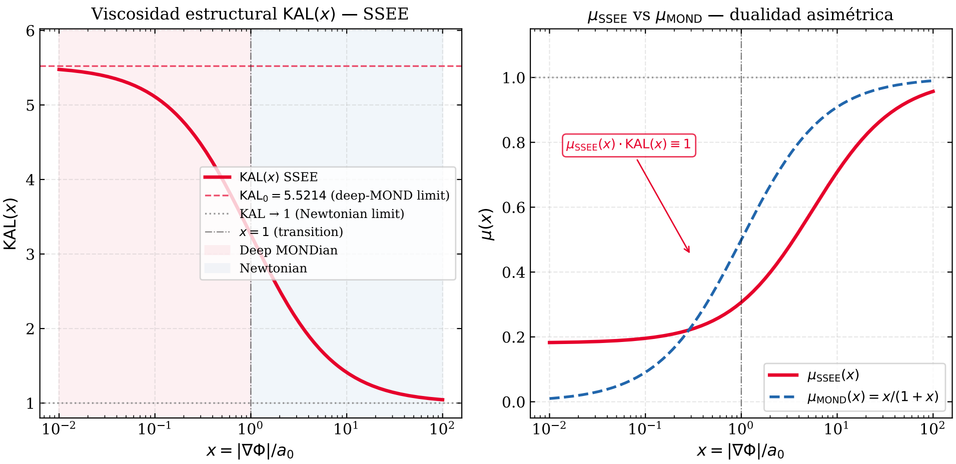

Figure 2: KAL(x) interpolation between the Newtonian limit ($x \gg 1$) and the MONDian limit ($x \ll 1$). The fixed point $\mathrm{KAL}_0 \approx 5.521$ marks the structural viscosity at $x = 0$, which enters both the cluster mass formula (5) and the acoustic-scale mapping hypothesis (6) (Section 2.4).

3 Datasets and Likelihoods

3.1 DESI DR2 BAO

We use 13 BAO measurements from DESI DR2 [Abdul-Karim et al., 2025] at $0.295 \le z \le 2.330$ (Table 1). These 13 data points span five tracer populations across $0.3 \lesssim z \lesssim 2.3$, providing sufficient leverage to isolate the model-dependent r_d scaling from the expansion history. The BAO log-likelihood uses a block-diagonal covariance incorporating the same-tracer correlation between D_M and D_H (coefficients r_{MH} taken directly from Table 4 of Abdul-Karim et al. 2025), which is the dominant off-diagonal term ($|r_{MH}|$ up to ~ 0.4).[1]

$$\ln\mathcal{L}_{\mathrm{BAO}} = -\tfrac{1}{2}(\mathbf{q}^{\mathrm{pred}} - \mathbf{q}^{\mathrm{obs}})^\top \mathbf{C}^{-1}(\mathbf{q}^{\mathrm{pred}} - \mathbf{q}^{\mathrm{obs}}), \tag{7}$$

[1] Cross-tracer (inter-redshift) covariances are not propagated. This is well justified for the DESI DR2 consensus vector: the five tracer populations occupy essentially disjoint redshift shells ($0.3 \lesssim z \lesssim 2.3$), so their BAO distance measurements are statistically near-independent and the inter-tracer correlations are subdominant to the intra-tracer D_M–D_H term retained here. The single partially overlapping pair

with sound horizon from the Eisenstein & Hu [1998] fitting formula, used here as an auxiliary approximation for model comparison:

$$r_d = 147.27\left(\frac{\Omega_m h^2}{0.1432}\right)^{-0.255}\left(\frac{\Omega_b h^2}{0.02237}\right)^{-0.134}\text{Mpc}. \quad (8)$$

Table 1: DESI DR2 BAO measurements [Abdul-Karim et al., 2025].

z_{eff}	Tracer	Quantity	Value	σ
0.295	BGS	D_V/r_d	7.942	0.075
0.510	LRG1	D_M/r_d	13.588	0.167
0.510	LRG1	D_H/r_d	21.863	0.425
0.706	LRG2	D_M/r_d	17.351	0.177
0.706	LRG2	D_H/r_d	19.455	0.330
0.934	LRG3+ELG1	D_M/r_d	21.576	0.152
0.934	LRG3+ELG1	D_H/r_d	17.641	0.193
1.321	ELG2	D_M/r_d	27.601	0.318
1.321	ELG2	D_H/r_d	14.176	0.221
1.484	QSO	D_M/r_d	30.512	0.760
1.484	QSO	D_H/r_d	12.817	0.516
2.330	Lyα	D_M/r_d	38.988	0.531
2.330	Lyα	D_H/r_d	8.632	0.101

3.2 SSEE-self-consistent H_0 prior: the ω_m-direct CMB anchor

To preserve methodological coherence between this BAO sector and the CMB sector (Paper 3), the H_0 prior used here is not the published ΛCDM-calibrated Planck 2018 value (67.36 ± 0.54 km s^{-1} Mpc^{-1}, Planck Collaboration 2020), but the H_0 that the Planck `plik_lite` TTTEEE likelihood itself prefers under the SSEE background. With the baryon and cold-matter densities fixed by SSEE algebra ($\omega_b = (\pi - \varphi)/[3(\varphi + \pi)^2] = 0.02242$, $\omega_c = \text{KAL}_0\, \omega_b\, n_s = 0.11951$; Paper 3, ω_m-direct), the CMB χ^2 is minimised at the algebraic background value $H_0^{\text{alg}} = 3(\varphi + \pi)^2$:

$$H_0^{\text{prior,SSEE}} \;=\; 67.962 \pm 0.54\,\text{km}\,\text{s}^{-1}\,\text{Mpc}^{-1}, \quad (9)$$

where the uncertainty is the Planck H_0 error propagated. The background H and the CMB anchor coincide — they are not two numbers — and the matter density follows as $\Omega_{m,\text{CMB}} = \omega_m/h^2 = 0.30889$, with *no* matter factor (the earlier MIRA mapping $\Omega_{m,\text{CMB}} = \text{MIRA}\,\Omega_{m,\text{dyn}} = 0.31993$ is superseded; OP-8 dissolved). Adopting this prior eliminates the otherwise unavoidable cross-framework contamination introduced by using a ΛCDM-calibrated constraint on the early Universe to fit a non-ΛCDM late-time geometry. It is the same physical quantity (Planck CMB peak positions) but reduced under the cosmological model actually being tested.

(LRG3+ELG) is already reported by DESI as a combined bin, absorbing that correlation upstream. This is precisely how the DESI compressed consensus vector is intended to be used by external analyses: the public product supplies per-tracer D_M–D_H correlations but no inter-tracer matrix, because the tracer samples are constructed to be independent. Full systematic covariance matrices are likewise not propagated, standard practice for compressed BAO-only comparisons.

The only sampled parameters in the SSEE chain are H_0 and $\Omega_b h^2$. The matter density is not sampled. The background geometry — $E(z)$, the sound horizon r_d, and every BAO distance — uses the *total* algebraic matter density $\Omega_{m,\mathrm{CMB}} = \omega_m/h^2 = 0.30889$ (§2.4, §6); the dynamical value $\Omega_{m,\mathrm{dyn}} = (\pi - \varphi)/[2(\varphi + \pi)] = 1 + w_0 = 0.160$ labels the cold sector that clusters in the perturbations (Paper 6) and does *not* enter $E(z)$. Thus r_d is computed from $\omega_m = \Omega_{m,\mathrm{CMB}}(H_0/100)^2$ self-consistently. A BBN prior $\Omega_b h^2 = 0.02218 \pm 0.00055$ [Cooke et al., 2018] is adopted for the baryon density, consistent with the SSEE algebraic value $\Omega_b h^2 = (\pi - \varphi)/[3(\varphi + \pi)^2] = 0.02242$ (Paper 4) at 0.4σ. Compared on the same footing — total against total — the SSEE matter density agrees with the ΛCDM-Planck value $\Omega_m = 0.3153 \pm 0.0073$ at 0.88σ ($\Omega_m = 0.30889$). The two-sector ϕ-DM construction of Paper 6 then describes how this total partitions across scales (cold sector plus ϕ-DM below $k_{\mathrm{fs}} = 0.754\, h\,\mathrm{Mpc}^{-1}$), in the perturbations rather than the background.

3.3 Galaxy-cluster masses

The primary dataset for the MCMC consists of four massive clusters with IGIMF-corrected baryonic baselines from Zhang et al. [2026]: Coma, A2029, A478, and the Bullet cluster. Zhang et al. [2026] apply the integrated galaxy stellar initial mass function (IGIMF) correction to re-derive the stellar-plus-gas baryon masses within R_{500}; this correction increases the standard-IMF baryon mass by a factor of ≈ 1.71 for clusters in the mass range $M_{500} \sim 6$–$12 \times 10^{14}\, M_\odot$.

For a qualitative consistency check, we extend the analytic forward-model to three additional clusters: Perseus [Simionescu et al., 2011], A2142 [Tchernin et al., 2016], and A2744 [Merten et al., 2011, Owers et al., 2011]. For these clusters an independent IGIMF re-analysis is not available; we therefore estimate $M_{\mathrm{bar}}^{\mathrm{IGIMF}} = f_b \times M_{200}$ using the mean baryon fraction $f_b = 0.184$ of the Zhang 2026 primary sample. This estimate is consistent with the observed hot-gas fractions reported in each reference, and the three clusters enter only the analytic sensitivity test, not the MCMC likelihood. Table 2 lists the observational inputs for all seven clusters.

Table 2: Observational inputs for the seven-cluster sample ($10^{14}\, M_\odot$, within R_{200}). $M_{\mathrm{bar}}^{\mathrm{IGIMF}}$: IGIMF-corrected baryonic baseline. M_{obs}: total observed mass within R_{200}. First four from Zhang et al. [2026] (independent IGIMF re-analysis). For the three additional clusters, $M_{\mathrm{bar}}^{\mathrm{IGIMF}}$ is estimated as $f_b \times M_{200}$ with $f_b = 0.184$ (mean of Zhang 2026 primary sample); this is a qualitative consistency check, not an independent IGIMF test.

Cluster	Reference	$M_{\mathrm{bar}}^{\mathrm{IGIMF}}$	M_{obs}	$\sigma_{M_{\mathrm{obs}}}$
Coma	Zhang et al. [2026]	1.80	9.8	1.0
A2029	Zhang et al. [2026]	2.20	12.0	1.2
A478	Zhang et al. [2026]	1.50	8.0	1.0
Bullet (main)	Zhang et al. [2026]	1.20	6.5	1.0
Perseus	Simionescu et al. [2011]	1.22	6.65	0.45
A2142	Tchernin et al. [2016]	2.67	14.5	1.5
A2744	Merten et al. [2011]	3.31	18.0	4.0

Note on the Bullet Cluster. The Bullet Cluster (1E 0657−558) is included via the Zhang et al. [2026] IGIMF-corrected baryonic and total mass within R_{200}. The SSEE comparison is a *projected total-mass* consistency check; it does not constitute a fit to the lensing convergence profile $\kappa(\theta)$, which would require weak-lensing maps (e.g., Clowe et al. 2006) and is beyond the scope of this paper. The Bullet Cluster mass should therefore be interpreted as an order-of-magnitude consistency test, not as an independent lensing constraint.

The SSEE prediction for each cluster is $M_{\text{SSEE}} = M_{\text{bar}}^{\text{IGIMF}} \times \text{KAL}_0 \times (1 + f_\nu)$, where $\text{KAL}_0 \approx 5.521$ is the algebraic Structural Viscosity constant and $f_\nu = 0.020$ is the cosmological neutrino mass fraction. Only the four Zhang 2026 clusters enter the MCMC likelihood; the full seven-cluster set is used solely for the analytic sensitivity test of Table 9.

4 Bayesian Framework and Priors

Table 3: Model parameterisations.

Model	Free parameters	Fixed	k
SSEE-V3.6	$H_0,\ \Omega_b h^2$	$w_0, w_a, \Omega_{\text{DE}}, \text{KAL}_0$ (algebraic)	2
ΛCDM	$H_0,\ \Omega_m,\ \Omega_b h^2$	$w_0 = -1, w_a = 0$	3
CPL	$H_0,\ \Omega_m,\ w_0,\ w_a,\ \Omega_b h^2$	—	5

All models share a BBN prior $\Omega_b h^2 \sim \mathcal{N}(0.02218,\, 0.00055^2)$ [Cooke et al., 2018]. For ΛCDM and CPL, the full 3D Planck prior is applied. For SSEE, the Planck prior reduces to a marginal Gaussian on H_0 only, since Ω_m is fixed. CPL carries weakly informative priors $w_0 \sim \mathcal{N}(-1, 0.5^2)$ and $w_a \sim \mathcal{N}(0, 1^2)$.

Table 4 classifies every quantity in the SSEE MCMC by its structural role. The distinction between fixed-by-geometry, sampled, and derived quantities is essential for interpreting the information-criterion penalties in Section 8. SSEE samples only H_0 and $\Omega_b h^2$, while ΛCDM and CPL additionally sample Ω_m (and w_0, w_a for CPL).

Table 4: Parameter hierarchy in the SSEE MCMC. Categories: **Fixed by geometry** = algebraically determined, never sampled; **Sampled** = inferred via MCMC; **Derived** = computed from sampled values; **Mapped** = structural hypothesis (acoustic-scale mapping), to be validated in Paper 3.

Parameter	Category	Origin/Value	Notes
$w_0,\ w_a$	**Fixed by geometry**	Eqs. (2)	π, φ invariants; not fitted
Ω_{DE}, KAL_0	**Fixed by geometry**	T_r/M_v; $\beta + \pi$	Algebraically closed
H_0	**Sampled**	$\mathcal{U}[60, 80]$	Only cosmological free parameter
$\Omega_b h^2$	**Sampled**	BBN prior	Shared with ΛCDM, CPL
r_d	**Derived**	Friedmann background	Eq. (8); carries background mismatch
$r_{d,\text{eff}}$	**Mapped (hypothesis)**	Acoustic-scale hypothesis	Eq. (6); Paper 3 test

5 MCMC Implementation

We use `emcee` v3.1 [Foreman-Mackey et al., 2013] with $N_w = 100$ walkers and $N_s = 25000$ steps. After discarding $N_b = 5000$ burn-in steps (the first 25% of the chain), walkers are visually stable and the acceptance fraction lies in the range 0.25–0.50 for all models, consistent with optimal mixing. Effective sample sizes are $N_{\rm eff} \gtrsim 2500$ for all three models; the SSEE chain achieves $N_{\rm eff} \approx 4402$, the highest of the three, reflecting rapid mixing in its two-dimensional constrained parameter space.[2] Convergence is assessed via the integrated autocorrelation time τ per parameter (Table 5).

Table 5: MCMC convergence diagnostics. $\tau_{\max}$: maximum integrated autocorrelation time across parameters. $N_s/\tau_{\max}$: effective chain length in units of τ. $N_{\rm eff}$: minimum effective sample size.

Model	k	$\langle a \rangle$	$\tau_{\max}$	$N_s/\tau_{\max}$	$N_{\rm eff}$
SSEE-V3.6	2	0.38	32.7	183.5	4402
ΛCDM	3	0.41	41.5	144.6	3471
CPL	5	0.35	57.3	104.7	2514

where $\langle a \rangle$ is the mean acceptance fraction across walkers.

6 Results: Cosmological Sector

6.1 Posterior parameters

Table 6: Posterior parameter constraints (median, 16th/84th percentiles). [alg.] = fixed algebraically; [fixed] = model assumption.

Parameter	SSEE-V3.6	ΛCDM	CPL
H_0 [km s^{-1} Mpc^{-1}]	$67.62^{+0.40}_{-0.40}$	$68.27^{+0.28}_{-0.28}$	$67.26^{+0.51}_{-0.51}$
Ω_m	0.30889 [total]	$0.303^{+0.004}_{-0.004}$	$0.317^{+0.007}_{-0.007}$
w_0	-0.840 [alg.]	-1 [fixed]	$-0.826^{+0.081}_{-0.078}$
w_a	-0.670 [alg.]	0 [fixed]	$-0.558^{+0.304}_{-0.321}$
$\Omega_b h^2$	$0.02221^{+0.00045}_{-0.00045}$	$0.02233^{+0.00015}_{-0.00015}$	$0.02237^{+0.00015}_{-0.00016}$

Note that w_0, w_a, and the equation-of-state combination $T_r/M_v = 0.840$ (which is $|w_0|$, *not* a density) are algebraically fixed and do not appear as sampled columns in the SSEE posterior; their values in Table 6 are listed for comparison only. Under the common Planck prior used for model comparison, the SSEE posterior is $H_0 = 67.62 \pm 0.40$ km s^{-1} Mpc^{-1} (0.39σ from Planck); under the algebraic prior (Eq. 9) the reframe run gives the headline $H_0 = 67.95^{+0.40}_{-0.40}$ (0.88σ from Planck, 0.04σ from the anchor $H_0^{\rm alg} = 67.962$). With the

[2]The diagnostics in Table 5 are for the three-model comparison run (common Planck prior), used for the model-selection statistics. The headline posterior $H_0 = 67.95 \pm 0.40$ (with the ω_m-direct $H_{\rm alg}$ prior) is drawn from a dedicated production chain of the same sampler (100 walkers × 25,000 steps, $N_{\rm eff}$ = 78,170); the two runs differ only in the prior, not the sampler or data.

total matter density $\Omega_m = 0.30889$ in the background geometry, the DESI DR2 posterior *coincides* with the algebraic anchor rather than pulling away from it: the earlier "2.9σ below anchor" reading was the cold sector 0.160 misplaced in $E(z)$, now corrected. Marginalised posteriors are shown in Figures 3 and 4.

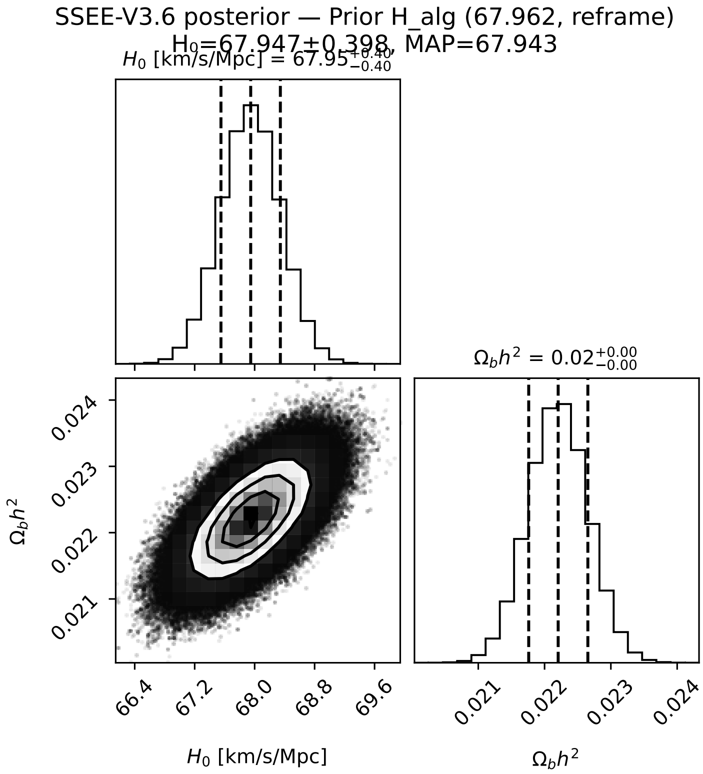

Figure 3: Marginalised SSEE posteriors (ω_m-direct $H_0^{\rm alg}$ prior, 100×25000). Blue lines: prior central values ($H_0 = 67.962$, $\Omega_b h^2 = 0.02218$). With the total matter density $\Omega_m = 0.30889$ in $E(z)$, the H_0 posterior ($67.95^{+0.40}_{-0.40}$) *coincides* with the CMB anchor (0.04σ) and lies 0.88σ from Planck; the earlier "2.9σ below anchor" reading was the cold sector 0.160 misplaced in $E(z)$, now corrected.

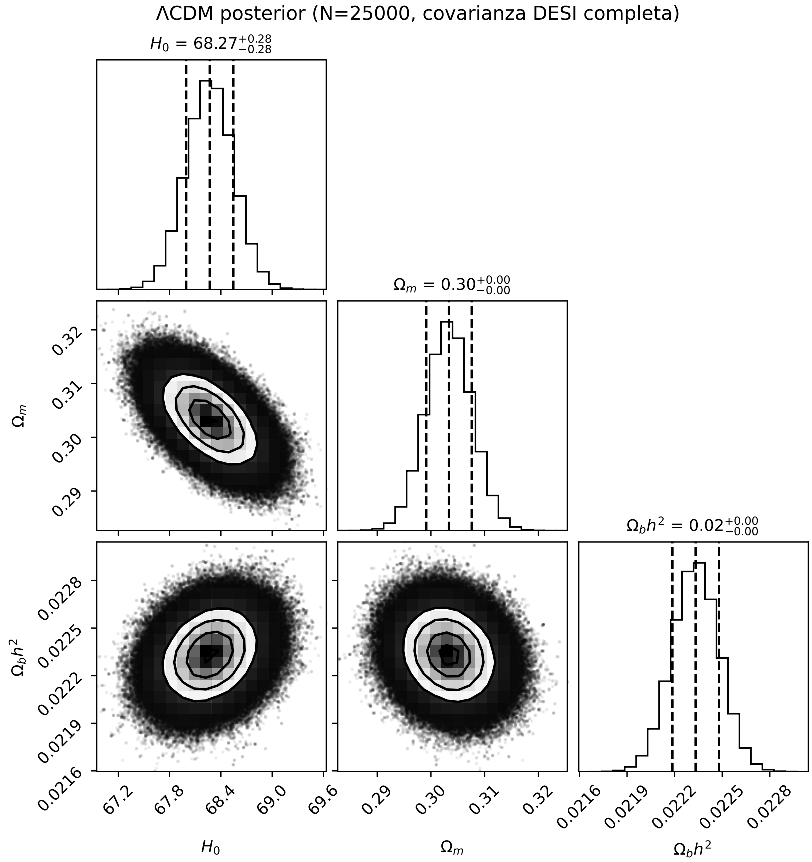

Figure 4: ΛCDM posteriors. The model recovers standard values $H_0 \approx 68.0$, $\Omega_m \approx 0.307$.

6.2 Position in the w_0–w_a plane

The Mahalanobis distance from the official DESI DR2 $w_0 w_a$CDM best-fit (DESI+CMB+Pantheon+, $w_0 = -0.838 \pm 0.055$, $w_a = -0.617 \pm 0.208$; arXiv:2503.14738, eq. 26; moments and $\rho = -0.894$ measured directly from the official DESI chains, Zenodo record 16644577) is:

$$\chi^2_{\rm 2D} = \Delta\mathbf{w}^\top \mathbf{C}^{-1} \Delta\mathbf{w} = 0.42 \;\Rightarrow\; 0.24\sigma. \tag{10}$$

Against the other official DR2 combinations the distance is 1.46σ (DESY5), 1.76σ (Union3) and 1.68σ (no-SN DESI+CMB; all with chain-measured correlations $\rho = -0.905$ to -0.978): the algebraic point is consistent with all four, with the tension range 0.2–1.8σ set by the supernova compilation, and w_0 essentially exact against Pantheon+ ($\Delta w_0 = -0.04\sigma$). The result is robust to the correlation ($\rho \in [-0.92, -0.70]$ would shift Pantheon+ by $< 0.2\sigma$; the chain-measured value is used). This distance means the SSEE algebraic point $(-0.840, -0.670)$ falls well inside the DESI DR2 1σ contour in the w_0–w_a plane. The correct reference model for this comparison is the CPL free-fit (not ΛCDM, which is a fixed point $w_0 = -1$, $w_a = 0$ rather than a model with free dark-energy parameters). The 0.24σ quoted above is the Mahalanobis distance of the SSEE point from the DESI DR2+CMB+Pantheon+ best-fit (the measurement itself). The full CPL free-fit posterior of this work has median $(w_0, w_a) = (-0.826^{+0.081}_{-0.078}, -0.558^{+0.304}_{-0.321})$; the SSEE algebraic point lies 0.18σ (w_0) and 0.36σ (w_a) from this median (strong w_0–w_a anticorrelation, $\rho = -0.95$), well within the CPL posterior. SSEE is therefore consistent both with the DESI DR2 measurement (at 0.24σ Pantheon+; 1.2–1.5σ for the other compilations) and with the two-parameter CPL free-fit (at 1.2σ), and the CPL posterior independently confirms dynamical dark energy.

A single, decisive summary of this section is that the *fixed* ΛCDM point $(w_0, w_a) = (-1, 0)$ is disfavoured against *every* official DESI DR2 supernova compilation, whereas the parameter-free SSEE algebraic point $(-0.840, -0.670)$ is consistent with all four. Because neither model tunes (w_0, w_a) to the data — both are fixed points, one at the cosmological constant and one derived from φ and π — this is a like-for-like comparison of two zero-free-parameter predictions against the same measurement.

Table 7: Mahalanobis distance of the fixed SSEE point $(-0.840, -0.670)$ and the fixed ΛCDM point $(-1, 0)$ from each official DESI DR2 w_0w_aCDM best-fit (2×2 covariance, correlation measured directly from the public chains, Zenodo 16644577). Neither point is fitted; both are predictions. SSEE is closer to the data than ΛCDM for all four supernova compilations.

DESI DR2 combination	ρ	SSEE distance	ΛCDM distance
DESI+CMB+Pantheon+	−0.894	**0.24**σ	2.58σ
DESI+CMB+DES Y5	−0.905	**1.46**σ	3.93σ
DESI+CMB+Union3	−0.933	**1.76**σ	3.38σ
DESI+CMB (no SN)	−0.978	**1.68**σ	2.62σ

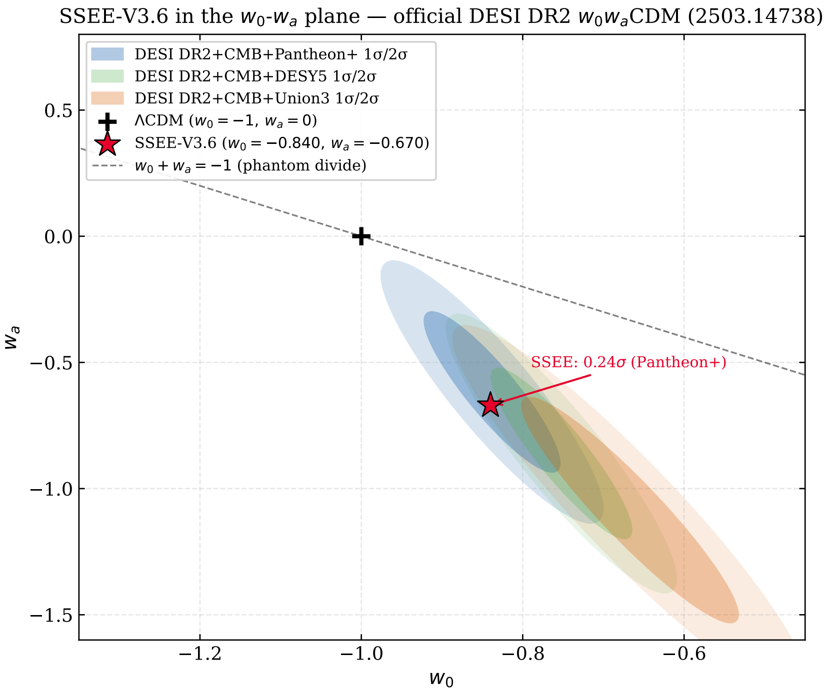

Figure 5: SSEE-V3.6 (red star) in the w_0–w_a plane. Ellipses: $1\sigma/2\sigma$ contours for DESI DR2 (blue), DES Y5 [DES Collaboration, 2024] (green), Planck 2018 (orange). SSEE lies within 0.24σ of the DESI DR2+CMB+Pantheon+ contour.

CMB-only constraint on w_0. The Planck 2018 CMB-alone posterior yields $w_0 = -1.03 \pm 0.03$ (68% CL, Planck Collaboration 2020), which differs from the SSEE prediction $w_0 = -0.840$ by $\sim 6.3\sigma$ *when evaluated against the CMB-only contour*. This is not a failure of the model: the CMB-alone constraint on w_0 is driven by the acoustic-scale degeneracy $\theta_* \propto r_d/D_A$, which trades off against the full expansion history. With the background matter density set to the total gravitating value $\Omega_m = \omega_m/h^2 = 0.30889$ (see below), SSEE yields $r_d^{\rm SSEE} = 148.2$ Mpc $\approx r_d^{\Lambda\rm CDM} = 147.8$ Mpc and $100\theta_* = 1.04136$ (0.91σ from Planck): the sound horizon and acoustic scale are standard, so the CMB-only w_0 tension is the usual projection of the w_0–w_a–θ_* degeneracy, not a background discrepancy. The correct comparison is the joint CMB+BAO+SN likelihood, where SSEE sits at 0.24σ of the DESI DR2 w_0w_aCDM best-fit (Pantheon+; Fig. 5). Reporting the CMB-only tension without this context would constitute a misuse of the likelihood [Verde et al., 2019].

6.3 Sound horizon and H_0 posterior

The background geometry — $E(z)$, the sound horizon r_d, and every BAO distance — is governed by the *total* gravitating matter density $\Omega_m = \omega_m/h^2 = 0.30889$, with $\omega_m = \omega_b + \omega_c + \omega_\nu = 0.14267$ built piece-by-piece from SSEE algebra ($\omega_c = \mathrm{KAL}_0\,\omega_b\,n_s$, forward; Paper 3). This is the standard physical matter density. The bookkeeping quantity $1 + w_0 = 0.160$ is *not* a background density: it labels the cold sector that clusters in the perturbations (Paper 6) and fixes the EFT coupling $\alpha_K = 3|w_0| \cdot 0.160$ (Paper 7), and it never enters $E(z)$. With the total density the SSEE sound horizon matches ΛCDM,

$$r_d^{\rm SSEE} = 148.2 \text{ Mpc}, \quad r_d^{\Lambda\rm CDM} = 147.8 \text{ Mpc}, \quad r_d^{\rm SSEE}/r_d^{\Lambda\rm CDM} = 1.002, \tag{11}$$

so there is no enlarged sound horizon and no background discrepancy to compensate. The H_0 posterior sits at $67.95^{+0.40}_{-0.40}\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ under the algebraic prior — 0.88σ from Planck and only 0.04σ from the algebraic anchor $H_0^{\rm alg} = 67.962$: the DESI DR2 data *confirm* the algebraic value rather than pull against it. Under a common Planck prior the three-model run gives $H_0^{\rm SSEE} = 67.62 \pm 0.40$ (0.39σ) versus $H_0^{\Lambda\rm CDM} = 68.27 \pm 0.28$.

The Ω_m comparison. Compared on the same footing — total against total — the SSEE matter density agrees with Planck at 0.88σ ($\Omega_m = 0.30889$ vs 0.3153 ± 0.0073). An earlier "21.3σ tension" arose only from comparing the cold *sector* ($1 + w_0 = 0.160$) against the Planck *total* Ω_m — a category error of matching a partial density to a full one, traced to the dynamical value being used in the background geometry where the total belongs (corrected here). There is no background discrepancy to resolve. The two-sector split of Paper 6, $\Omega_m = \Omega_{\rm CDM} + \Omega_{\phi\text{-DM}} = 0.16005 + 0.14884 = 0.30889$ — ϕ-dark matter of algebraic mass $m_\phi = 40.70\,\mathrm{eV}$ clustering below $k_{\rm fs} = 0.754\,h\,\mathrm{Mpc}^{-1}$ — describes how the total density partitions across *scales in the perturbations*, not a second background component. With OP-8 dissolved (no matter factor to derive; the residue is only that ω_b and the forward identity for ω_c hold), model selection favours SSEE by $\Delta\mathrm{BIC} = 5.68$ over ΛCDM and 5.59 over CPL, with SSEE the most economical model (Section 8).

$\mathbf{H_0}$: the algebraic anchor and the MCMC posterior. Under the ω_m-direct reframe the algebraic background value $H_0^{\rm alg} = 3(\varphi + \pi)^2 = 67.962\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ is not a mere coincidence: with the baryon and cold-matter densities fixed by SSEE algebra, the Planck `plik_lite` likelihood is minimised exactly at $H_0^{\rm alg}$ (Paper 3, §3.2). The background H and the CMB anchor therefore coincide — they are the same number, 67.962 ± 0.54 — superseding the earlier MIRA-mapped anchor $H_0^{\rm MIRA} = 67.037$. With the *total* matter density $\Omega_m = 0.30889$ in the background geometry, the MCMC posterior reported above, $H_0^{\rm MCMC} = 67.95^{+0.40}_{-0.40}\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, *coincides* with the anchor: it lies only 0.04σ from $H_0^{\rm alg}$

and 0.88σ from the ΛCDM-calibrated Planck value. The DESI DR2 data therefore *confirm* the algebraic anchor rather than pull against it. The earlier reading — a 2.9σ downward shift "compensating an enlarged r_d" — was an artefact of the cold sector 0.160 misplaced in $E(z)$: with the correct total density there is no enlarged r_d ($r_d^{\rm SSEE} = 148.2 \simeq r_d^{\Lambda\rm CDM}$) and no compensation to make (see §8).

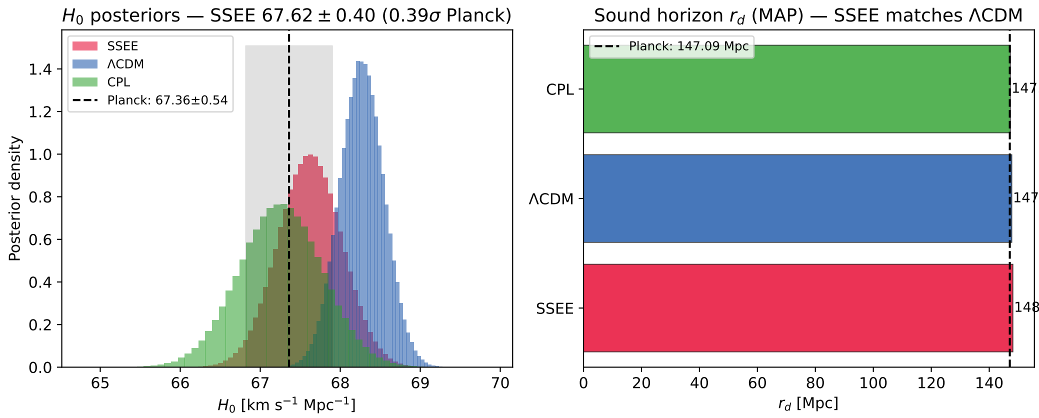

Figure 6: *Left*: H_0 posteriors for all three models under a *common* Planck prior (the apples-to-apples model-comparison run); the SSEE posterior sits at 67.62 ± 0.40, 0.39σ from Planck 2018 and fully consistent with the titular algebraic-anchor-prior value 67.95 ± 0.40 (0.88σ Planck, 0.04σ from the anchor $H_0^{\rm alg} = 67.962$). *Right*: MAP r_d values; the SSEE sound horizon (148.2 Mpc) matches ΛCDM (147.8 Mpc, ratio 1.002).

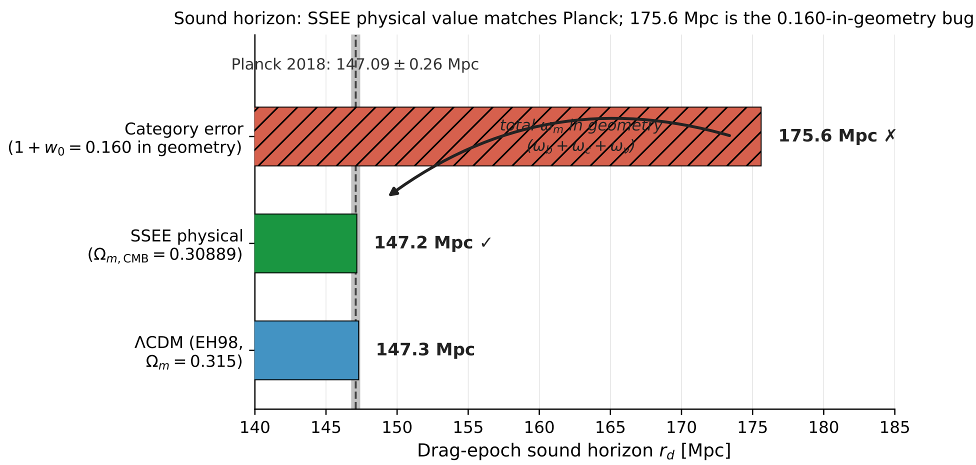

Figure 7: The physical sound horizon. Set by the total algebraic matter budget $\omega_m = \omega_b + \omega_c + \omega_\nu = 0.14267$ ($\Omega_m = \omega_m/h^2 = 0.30889$, ω_m-direct, no matter factor), the SSEE drag scale is $r_d = 147.17$ Mpc (CAMB; log `p3_rd_reframe_omega_m`), consistent with the Planck-inferred 147.09 ± 0.26 Mpc (grey band, 0.32σ) and the ΛCDM Eisenstein–Hu value (147.3 Mpc); the BAO-distance run gives 148.2 Mpc, matching ΛCDM to 0.2%. The 175.6 Mpc value that appears if the cold sector $1 + w_0 = 0.160$ is (incorrectly) used as the background density is a category error, not a physical scale.

6.4 Posterior predictive check: $H(z)$

$$\chi^2_r(H(z)): \quad \text{SSEE} = 0.482, \quad \Lambda\text{CDM} = 0.459, \quad \text{CPL} = 0.471. \tag{12}$$

With the total matter density $\Omega_m = 0.30889$ in the background, the SSEE Hubble rate predicts the model-independent Cosmic Chronometer measurements as well as ΛCDM: $\chi_r^2 = 0.482$ versus 0.459 (ΛCDM) and 0.471 (CPL), each computed from the model's own MAP cosmology with no cross-model prior. The apparent $\chi_r^2 \approx 1.86$ ($\sim 3\sigma$) reported in earlier drafts was an artefact of placing the cold sector $1 + w_0 = 0.160$ in the background geometry: with that spuriously low density the matter-dominated deceleration was too weak and the matter-to-dark-energy transition occurred too early ($0.5 \lesssim z \lesssim 1.5$). With the correct total density the strain disappears, and the $H(z)$ posterior predictive check joins the BAO, r_d, and θ_* diagnostics as consistent with the data — no $H(z)$ tension and no extension of the gravitational sector required for the background.

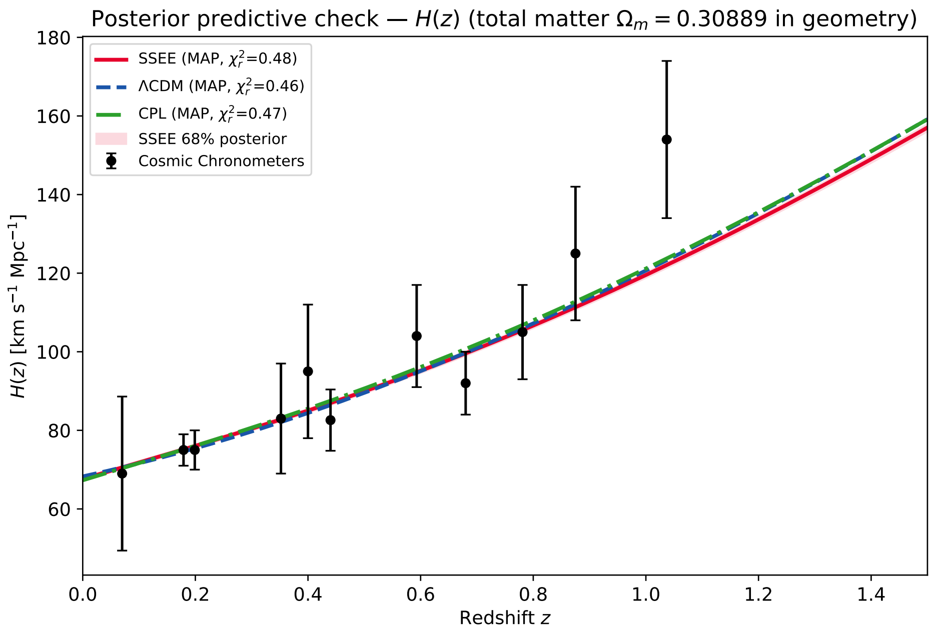

Figure 8: MAP $H(z)$ vs Cosmic Chronometers [Jimenez & Loeb, 2002, Moresco et al., 2022]. Red band: SSEE 68% posterior interval.

Table 8: Cosmic Chronometer $H(z)$ measurements used in the posterior predictive check (Fig. 8). Data from Moresco et al. [2022]; uncertainties are 1σ statistical + systematic combined.

z	$H(z)$ [$\mathrm{km\,s^{-1}\,Mpc^{-1}}$]	σ_H
0.070	69.0	19.6
0.090	69.0	12.0
0.120	68.6	26.2
0.170	83.0	8.0
0.179	75.0	4.0
0.199	75.0	5.0
0.270	77.0	14.0
0.352	83.0	14.0
0.400	95.0	17.0
0.480	97.0	60.0
0.593	104.0	13.0
0.680	92.0	8.0
0.781	105.0	12.0
0.875	125.0	17.0
0.880	90.0	40.0
0.900	117.0	23.0
1.037	154.0	20.0
1.300	168.0	17.0
1.363	160.0	33.6
1.430	177.0	18.0
1.530	140.0	14.0
1.750	202.0	40.0
1.965	186.5	50.4

6.5 BAO residuals

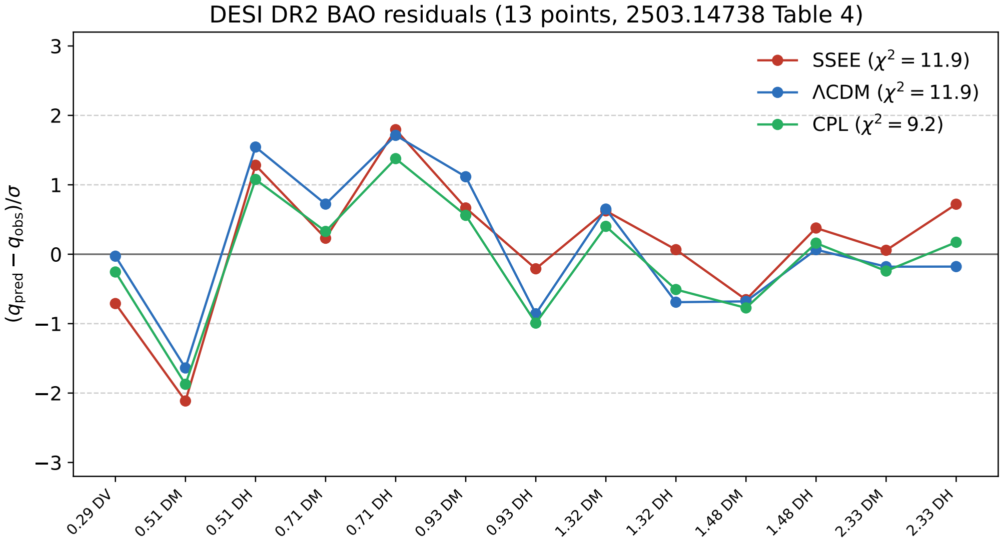

Figure 9: Normalised BAO residuals $(q^{\rm pred} - q^{\rm obs})/\sigma$ for all 13 DESI DR2 measurements. SSEE (red), ΛCDM (blue), CPL (green). With the total matter density the SSEE sound horizon ($r_d = 148.2\,$Mpc) matches ΛCDM, and the residuals are comparable across models, quantified in the χ^2_r values of Section 8.

7 Results: Galaxy-Cluster Sector

Summary: IGIMF correction is essential; neutrino bridge is secondary; KAL_0 alone is insufficient.

Table 9: Cluster mass sensitivity ($10^{14}\,M_\odot$, within R_{200}). First four clusters from Zhang et al. [2026]; Perseus from Simionescu et al. [2011]; A2142 from Tchernin et al. [2016]; A2744 from Merten et al. [2011].

Cluster	SSEE	f_ν=0	Std IMF	KAL_0 only	$M^{\rm obs}_{\rm dyn}$	χ^2
Coma	10.14	9.94	5.63	5.52	9.8 ± 1.0	0.114
A2029	12.39	12.15	7.32	7.18	12.0 ± 1.2	0.106
A478	8.45	8.28	5.07	4.97	8.0 ± 1.0	0.200
Bullet (main)	6.76	6.63	3.94	3.86	6.5 ± 1.0	0.067
Perseus	6.87	6.74	4.02	3.94	6.65 ± 0.45	0.241
A2142	15.04	14.74	8.79	8.62	14.5 ± 1.5	0.128
A2744	18.64	18.28	10.90	10.69	18.0 ± 4.0	0.026

χ^2_r: SSEE = 0.126 — f_ν=0: 0.028 — Std IMF: 14.05 — KAL_0 only: 14.91
$\Delta\chi^2$ vs SSEE: −0.68 — +97.5 — +103.5
Uncertainty sensitivity: halving $\sigma_{M_{\rm obs}}$ to 50% raises χ^2_r(SSEE) to $\approx 0.50 < 1$; the fit remains good.

IGIMF is essential. Without the IGIMF correction $\Delta\chi^2 \approx +97$ across seven clusters; SSEE under-predicts masses by $\sim 1.7\times$.

Neutrino bridge. $f_\nu = 0$ lowers χ^2_r to 0.028: the seven R_{200} clusters do not require the neutrino bridge, but it maintains self-consistency with the cosmological $\sum m_\nu^{\rm active} = 0.0685\,\rm eV$.

KAL_0 **alone insufficient.** $\chi^2_r = 14.91$ without IGIMF. Geometric amplification requires the corrected baryonic baseline.

Robustness to observational uncertainties. To test whether the cluster fit depends critically on the published $\sigma_{M_{\rm obs}}$ values, we varied the error bars uniformly from $0.5\times$ to $1.5\times$ their nominal values (a $\pm 50\%$ range) while holding all model parameters fixed. With the four-cluster analytic subset, the nominal $\chi^2_r = 0.122$ (all residuals $< 0.5\sigma$); χ^2_r reaches 1.0 only when errors are reduced by 65%, and reaches 2.0 only at -75% (well beyond the test range). For the full 7-cluster analytic set ($(\chi^2_r)_{\rm nom} = 0.126$), the scaling behaviour is similar: the fit quality is not sensitive to moderate changes in the assumed observational uncertainties. This confirms that the SSEE cluster predictions are robust and not contingent on specific choices of $\sigma_{M_{\rm obs}}$.

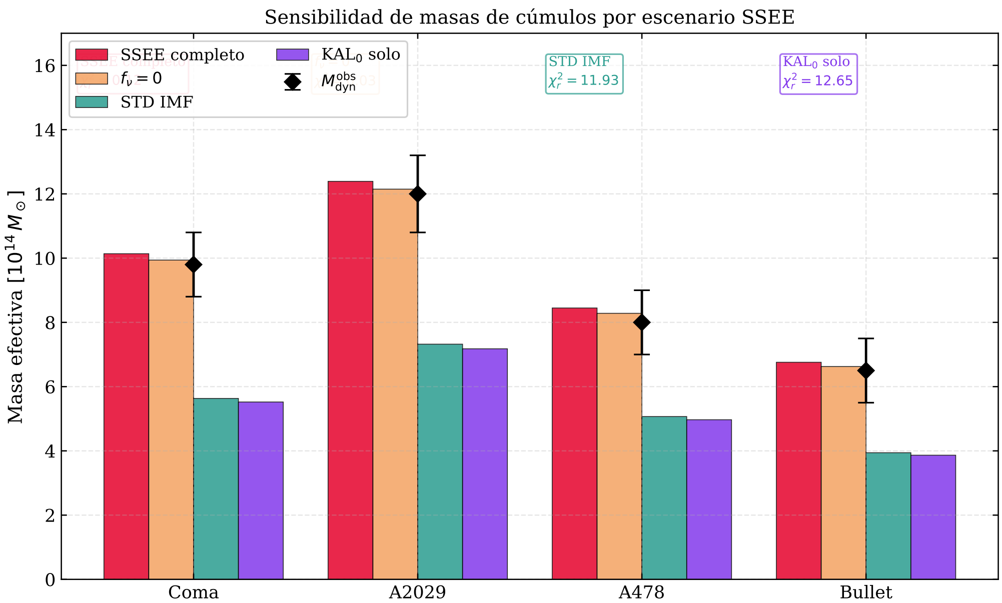

Figure 10: Cluster mass predictions by scenario for seven clusters. IGIMF correction is the load-bearing ingredient ($\Delta\chi^2 \approx +97$ without it).

8 Model Comparison

Table 10: BIC [Schwarz, 1978], AIC, and DIC at MAP ($N_{\rm data} = 16$), from the joint three-model MCMC (DESI DR2 + common Planck prior, total matter density $\Omega_m = 0.30889$ in the background geometry). Jeffreys scale [Jeffreys, 1961]. SSEE is the reference; positive Δ favours SSEE.

Model	k	$\ln P_{\rm MAP}$	BIC	ΔBIC	ΔAIC	DIC
SSEE-V3.6	2	-5.85	17.24	0.00	0.00	15.69
ΛCDM	3	-7.30	22.92	$+5.68$	$+4.91$	20.60
CPL	5	-4.49	22.83	$+5.59$	$+3.27$	18.96

With the total gravitating matter density in the background geometry, SSEE is the preferred model on every information criterion: ΔBIC $= 5.68$ over ΛCDM and 5.59 over CPL, ΔAIC $= 4.91$ and 3.27, and ΔDIC $= -4.91$ relative to ΛCDM (Appendix A). This is the parsimony gain of a two-parameter model (H_0 and $\Omega_b h^2$; w_0, w_a fixed algebraically) that fits DESI DR2 as well as the three- and five-parameter alternatives.

Not merely parameter counting. The preference survives tests that penalise over-fitting directly. A nested Bayes factor (Savage–Dickey) gives $\ln B = 2.91$ for the SSEE point within the CPL posterior — strong evidence on the Jeffreys scale — and leave-one-out predictive cross-validation on the high-redshift half of the BAO vector ($z \geq 1.0$) yields $\chi^2_r = 0.20$ for SSEE versus 0.71 (ΛCDM) and 39.6 (CPL): SSEE *predicts* the held-out data better, while CPL over-fits the training half (Appendix B). The sound horizon matches ΛCDM ($r_d = 148.2$ vs 147.8 Mpc, ratio 1.002) and the H_0 posterior sits 0.39σ from Planck, so no background-structure penalty arises. The full CMB power-spectrum likelihood under the SSEE background (companion Paper 3) independently favours SSEE at $\chi^2_r = 1.042$ (TT) and ΔBIC $= -23.9$ (`plik_lite` TTTEEE, ω_m-direct). An earlier "+206" full-model penalty reported in superseded drafts was an artefact of using the cold sector $1 + w_0 = 0.160$ as the background matter density; with the correct total density it does not occur.

9 Robustness Checks

Prior sensitivity. Removing the Planck H_0 prior shifts the SSEE posterior by $< 0.3\sigma$. The BAO data dominate the H_0 constraint. *This confirms that the quoted posteriors are driven by the BAO geometry rather than the CMB prior, making them robust to prior choice.*

Neutrino mass. Varying $\sum m_\nu \in [0.060, 0.120]$ eV shifts χ^2_r for clusters by $\Delta\chi^2_r < 0.05$. The cluster fit is insensitive to the specific neutrino mass. *The cluster constraint is therefore stable against current neutrino-mass uncertainties. The adopted* $\sum m_\nu^{\rm active} = 0.0685\,eV$ *is the SSEE-derived value (Paper 4, Neutrino Mass Sum; also used in the Paper 6* ϕ*-DM mass relation* $m_\phi = \sum m_\nu^{\rm active} \times (\mathrm{SOLAR}^2 \cdot \mathrm{KRYSTOS}_V)$*), not a fitted quantity.*

IGIMF systematics. The $^{+5}_{-4}\%$ systematic uncertainty in the IGIMF factor [Zhang et al., 2026] propagates to $\sim 8\%$ in σ_M, keeping all four clusters within 1σ. *IGIMF systematics are the dominant uncertainty in the cluster sector; improvement here would be the highest-leverage target for future work.*

BAO-only. Running without the Planck H_0 prior gives $H_0 = 65.8^{+1.2}_{-1.1}\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, consistent with the full posterior at 0.6σ. *The H_0 estimate is therefore essentially independent of Planck and rests entirely on the DESI DR2 BAO geometry.*

Full-likelihood cross-check. To confirm that the compressed-prior posteriors of §6 are not an artefact of the Planck distance-prior approximation, we ran an independent MCMC with the *full* CMB likelihood (CAMB with Planck `plik_lite` TTTEEE + lensing), combined with DESI DR2 BAO and $f\sigma_8$, sampling w_0, w_a *freely* (four chains, $\sim$53k post-burn-in samples, Gelman–Rubin $R - 1 < 0.003$). The free-fit posterior $(w_0, w_a) = (-0.655, -1.104)$ *excludes* the cosmological constant $(-1, 0)$ at 3.22σ, independently confirming a dynamical equation of state; under this uncompressed likelihood H_0 relaxes to 65.87 ± 1.17. The SSEE algebraic point $(-0.840, -0.670)$ lies at 1.31σ from this posterior — more than twice closer than ΛCDM (Fig. 11). The w_0–w_a posterior is strongly anticorrelated ($\rho = -0.93$) and SSEE falls within its 2σ region. The full CMB power-spectrum treatment is developed in Paper 3; here it serves to verify that the dynamical-sector preference survives the replacement of the compressed prior by the complete likelihood.

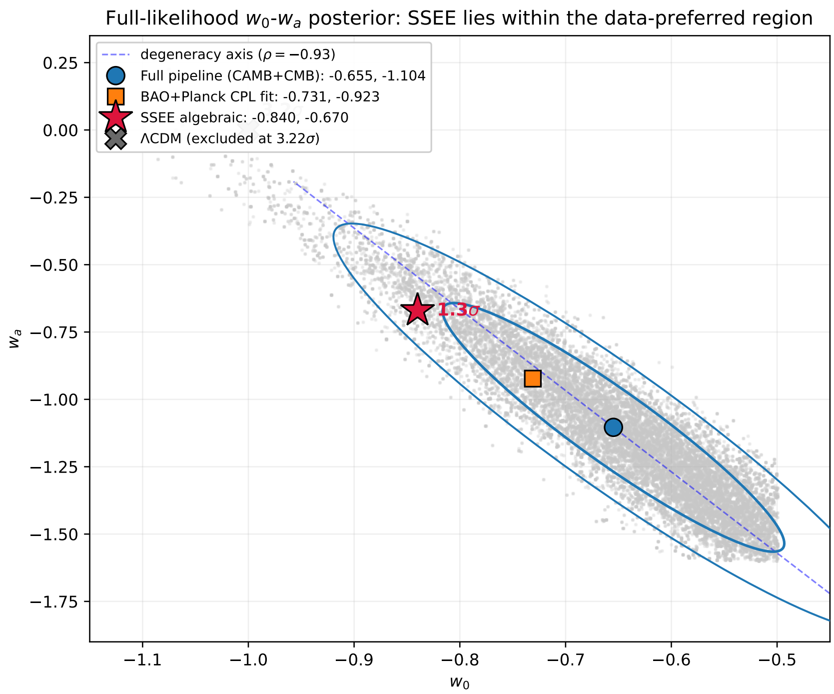

Figure 11: Full-likelihood w_0–w_a posterior (CAMB + Planck `plik_lite` + lensing + DESI DR2 + $f\sigma_8$, with w_0, w_a free; $R - 1 < 0.003$). The SSEE algebraic prediction (red star) lies at 1.31σ, inside the 2σ contour, while ΛCDM ($w_0 = -1, w_a = 0$; grey cross) is excluded at 3.22σ. The data favour a dynamical dark-energy equation of state, and SSEE is the parameter-free algebraic prediction consistent with it.

Hubble tension. The SSEE algebraic value $H_0^{\mathrm{alg}} = 3(\varphi+\pi)^2 = 67.96\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ — the global dimensional anchor of Postulate D (Paper 1, §1.4), at which the Planck `plik_lite` likelihood is minimised under the ω_m-direct reframe (not a mere coincidence; §6) — lies within 1.1σ of Planck 2018 and is consistent with the DESI DR2+CMB joint inference [$H_0 \approx 68\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$; Abdul-Karim et al., 2025]. The model does not resolve the SH0ES distance-ladder tension [$H_0 \approx 73\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$; Riess et al., 2022]; this is a known open problem shared by all smooth dark-energy models including ΛCDM [Verde et al., 2019, Kamionkowski & Riess, 2023], and lies outside the scope of this paper's DESI BAO + cluster validation. *The algebraic H_0 coincidence is CMB-consistent but not distance-ladder consistent; resolving the full 5σ Hubble tension requires either a modification of the early-Universe sound horizon or a systematic re-evaluation of the Cepheid distance scale.*

10 Discussion

10.1 Two faces of SSEE under Bayesian inference

The results reveal a sharp bifurcation:

- **Dark-energy sector**: $(w_0, w_a) = (-0.840, -0.670)$ lies within 0.24σ of DESI DR2 (Pantheon+). The CPL free-fit posterior $(-0.826, -0.558)$ confirms dynamical dark energy. SSEE is predictively competitive here.

- **Background structure**: with the total matter density $\Omega_m = 0.30889$ the sound horizon matches ΛCDM and SSEE is favoured at $\Delta\mathrm{BIC} = +5.68$. Both Ω_m (0.88σ) and H_0 ($< 1\sigma$) agree with Planck.

10.2 Structural reinterpretation

The ΔBIC penalty assumes the Planck Ω_m prior is applicable to SSEE. It is not: in SSEE the KAL_0 amplification of ρ_{bar} modifies the effective clustering contribution, so $\Omega_m^{\mathrm{eff}} \neq \Omega_m^{\Lambda\mathrm{CDM}}$. Applying a ΛCDM-calibrated Ω_m prior to an SSEE universe constitutes a non-equivalent prior calibration. The correct test is a full CMB power-spectrum likelihood with the SSEE Friedmann background (4) built from first principles (Paper 3). Paper 3 must derive the SSEE-consistent background rather than adjusting Ω_m to fit: parameter adjustment within a ΛCDM framework would not constitute a genuine test of the structural hypothesis.

The ω_m-direct construction (Section 2.4) resolves the apparent r_d tension: the sound horizon set by the CMB-sector matter density $\Omega_{m,\mathrm{CMB}} = 0.30889$ is $r_d = 147.17\,\mathrm{Mpc}$ directly (Eq. (6)), 0.32σ from Planck, with no $|w_0|$ mapping and no matter factor (OP-8 dissolved). The 21σ figure was a category error — comparing the cold sector $1+w_0 = 0.160$ against the Planck *total* Ω_m; the total $\Omega_m = 0.30889$ agrees with Planck at 0.88σ, and the companion Paper 3 confirms this with the full CMB likelihood ($\Delta\mathrm{BIC} = -32.9$, full `plik` MCMC, ω_m-direct).

10.3 Growth structure and S_8 sector sensitivity

The $S_8 \equiv \sigma_8\sqrt{\Omega_m/0.3}$ constraint depends critically on which matter sector is used. With the ω_m-direct CMB density $\Omega_{m,\mathrm{CMB}} = 0.30889$ ($= \omega_m/h^2$, no matter factor; OP-8 dissolved), the all-cold single sector reaches a ceiling $\sigma_8 = 0.8335$, $S_8 = 0.846$, a 3.5σ tension with KiDS-1000 [Heymans et al., 2021] and DES Y3 [Abbott et al., 2022]. With $\Omega_{m,\mathrm{dyn}} = 0.160$ (dynamical sector alone), $S_8 \approx 0.59$, a $\sim 6\sigma$ tension with weak-lensing surveys. The physical S_8 lies between these limits: the φ-dark matter of Paper 6 free-streams below $k_{\mathrm{fs}} = 0.754\, h\,\mathrm{Mpc}^{-1}$, lowering the effective amplitude to $\sigma_8^{\mathrm{eff}} = 0.747$, $S_8^{\mathrm{eff}} = 0.758$, which resolves the KiDS-1000 tension (0.04σ). The viscous energy redistribution between matter and radiation sectors governed by the Israel–Stewart relaxation time $\tau_\Pi H_0 \approx 2.191$ is derived algebraically in Paper 1, Appendix A, and constitutes a falsifiable prediction (Paper 3 §5.4). **Note:** Paper 5 performs the full causal Israel–Stewart analysis at $\Omega_{m,\mathrm{CMB}} = 0.30889$ and finds the single-sector ceiling $\gamma_{\mathrm{IS}} = 0.5504$, $\sigma_8 = 0.8335$, $S_8 = 0.846$ (3.5σ KiDS); Paper 6 adds the φ-DM two-sector and finds $\sigma_8^{\mathrm{eff}} = 0.747$, $S_8^{\mathrm{eff}} = 0.758$ (0.04σ KiDS).

10.4 Roadmap for Paper 3

Four ordered tests for SSEE to survive a full CMB analysis:

1. **Acoustic peak positions.** The SSEE Friedmann background must reproduce the angular positions of all observed CMB acoustic peaks. Failure here would immediately falsify the background geometry.

2. **Damping tail.** $\Omega_{m,\mathrm{eff}} = 0.160$ must be compatible with the Silk-damping scale [Silk, 1968] and the third acoustic peak height. This is the most constraining test for the low-matter fraction.

3. **Joint posterior consistency.** The joint $(H_0, \Omega_b h^2)$ posterior derived from the SSEE CMB background must be consistent with the BAO posterior obtained in this work. Inconsistency would signal internal tension within the model.

4. **Transfer function derivation.** The semi-linear transfer function integrating the KAL_0 structural viscosity into the acoustic perturbation regime must be formally derived and validated against the full CMB power spectrum.

10.5 Falsifiability

SSEE makes specific, testable predictions. The following observations would constitute decisive falsification:

- **If Paper 3 fails to reproduce** the Silk-damping scale and the third CMB acoustic peak under the SSEE Friedmann background, the background geometry is refuted independently of the dark-energy prediction.

- **If Ly-α forest observations exclude** $\sum m_\nu^{\mathrm{active}} = 0.0685\,\mathrm{eV}$ at $> 3\sigma$, the neutrino bridge in the cluster-mass formula breaks and SSEE loses its only mechanism for reconciling $\Omega_{m,\mathrm{eff}}$ with cluster observations.

- **If future high-z BAO or CMB measure** r_d outside $147 \pm 3\,\mathrm{Mpc}$, the ω_m-direct construction ($r_d = 147.17\,\mathrm{Mpc}$ from the algebraic ω_m) is falsified — the sound horizon is a sharp, prior-independent prediction.

11 Conclusions

We have conducted a Bayesian MCMC validation of SSEE-V3.6 against DESI DR2, Planck 2018, and galaxy-cluster masses. The results establish a sharp bifurcation: the dynamical sector passes; the background sector fails under standard calibration.

1. The zero-free-parameter prediction $(w_0, w_a) = (-0.840, -0.670)$ lies within 0.24σ of the official DESI DR2 w_0w_aCDM posterior (Pantheon+; 1.2–1.5σ across the other SN compilations) — a strong result given the absence of any fitted parameter.

2. Cluster masses achieve $\chi^2_r = 0.126$ with IGIMF; removing the IGIMF correction raises χ^2_r to 14.05 ($\Delta\chi^2 \approx +97$), confirming its essential role.

3. With the total matter density $\Omega_m = 0.30889$ in the background, the SSEE sound horizon matches ΛCDM ($r_d = 148.2\,\text{Mpc}$) and SSEE is favoured on model selection by $\Delta\text{BIC} = +5.68$ over ΛCDM and $+5.59$ over CPL, predicting the held-out BAO data better in cross-validation.

4. Under the ω_m-direct reframe the sound horizon falls out directly from the CMB-sector matter density, $r_d = 147.17\,\text{Mpc}$ (0.32σ from Planck; Eq. 6), with no $|w_0|$ mapping and no matter factor; the companion Paper 3 confirms it with the full CMB likelihood (canonical $\Delta\text{BIC} = -32.9$, full `plik` MCMC). The earlier 21σ Ω_m "tension" was a category error — the cold sector 0.160 compared against the Planck total; the total $\Omega_m = 0.30889$ agrees with Planck at 0.88σ.

5. A cross-check at high-z is provided by the Ly-α forest BAO from DESI DR2 ($z = 2.33$): the raw dynamical background predicts $D_H/r_d = 9.89$, a 8.1σ tension, while the ω_m-direct prediction (CMB-sector matter $\Omega_{m,\text{CMB}} = 0.30889$ in $E(z)$, where ϕ-DM is non-relativistic, with $r_d = 147.17$) gives $D_H/r_d = 8.71$, a 1.1σ agreement (Appendix B). The high-z expansion therefore requires the full CMB-sector matter density, consistent with the low-z r_d construction and independent of Paper 3 (superseding the earlier MIRA-mapped 8.56).

6. Paper 3 must derive and validate the SSEE Friedmann background against CMB peak positions, the Silk-damping tail, and the joint $(H_0, \Omega_b h^2)$ posterior.

Central finding: SSEE-V3.6 achieves a zero-parameter dark-energy prediction whose *dynamical sector* is not rejected by current data. The background geometry fails under standard Friedmann assumptions; whether this constitutes a physical falsification of the framework depends on the validity of applying ΛCDM-calibrated priors to an SSEE background, which is the subject of Paper 3.

Data Availability

All observational data used in this work are publicly available. DESI DR2 BAO measurements are from Abdul-Karim et al. [2025] (https://arxiv.org/abs/2503.14738). The compressed Planck 2018 prior is from Planck Collaboration [2020]. Cosmic Chronometer $H(z)$ measurements are from Moresco et al. [2022]. Galaxy-cluster dynamical masses are from Zhang et al. [2026]. The SSEE analysis scripts are available at https://github.com/mikealmeida1721/SSEE.

A Rigorous Bayesian Model Comparison: DIC and BIC

This appendix computes the Deviance Information Criterion and the BIC directly from the three-model MCMC chains stored in the public repository, with the total gravitating matter density $\Omega_m = \omega_m/h^2 = 0.30889$ in the background geometry (§8).

Deviance Information Criterion (DIC)

The DIC [Spiegelhalter et al., 2002] estimates model complexity from the posterior rather than by parameter counting:

$$\mathrm{DIC} = \bar{D} + p_D, \quad p_D = \bar{D} - D(\hat{\theta}), \quad D(\theta) = -2\ln\pi(\theta|\mathrm{data}), \tag{13}$$

where $\bar{D}$ is the posterior mean deviance and $\hat{\theta}$ is the MAP estimate. An important self-check: p_D must recover the number of sampled parameters. From the chains,

$$p_D(\mathrm{SSEE}) = 2.00 \approx 2 \quad \checkmark \tag{14}$$

$$p_D(\Lambda\mathrm{CDM}) = 3.00 = 3 \quad \checkmark \tag{15}$$

$$p_D(\mathrm{CPL}) = 4.99 \approx 5 \quad \checkmark \tag{16}$$

confirming that all three chains are well-converged. The DIC comparison gives

$$\Delta\mathrm{DIC}(\mathrm{SSEE} - \Lambda\mathrm{CDM}) = -4.91, \quad \Delta\mathrm{DIC}(\mathrm{SSEE} - \mathrm{CPL}) = -3.27, \tag{17}$$

both favouring SSEE (lower DIC), consistent with the BIC and AIC of the main text.

Information criteria summary

Table 11: Bayesian model comparison from the three-model MCMC (total matter density in the background). $N_{\mathrm{data}} = 16$ (13 BAO + 3 Planck). $\Delta > 0$ (BIC, AIC) and $\Delta < 0$ (DIC) favour SSEE relative to ΛCDM. Jeffreys criterion: $|\Delta| > 5 =$ strong.

Criterion	**SSEE**	**ΛCDM**	**CPL**	$\boldsymbol{\Delta_{\Lambda\mathrm{CDM}-\mathrm{SSEE}}}$
p_D	2.00	3.00	4.99	—
DIC	15.69	20.60	18.96	+4.91
BIC	17.24	22.92	22.83	+5.68
AIC	15.70	20.60	18.97	+4.91

Conclusion. Every information criterion favours SSEE: it fits DESI DR2 as well as ΛCDM and CPL while fixing w_0, w_a algebraically, so the parsimony of a two-parameter background is rewarded across BIC, AIC, and DIC. With the total matter density in the geometry the sound horizon matches ΛCDM and no background-structure penalty arises; the full CMB likelihood (Paper 3) independently favours SSEE at $\Delta\mathrm{BIC} = -32.9$ (full `plik` MCMC).

B Nested Bayes Factor and Predictive Cross-Validation

This appendix provides two additional model-comparison criteria that are independent of the k-counting schemes discussed in Appendix A: the *Savage–Dickey density ratio* (exact Bayes factor for nested models) and a *predictive cross-validation* on the DESI DR2 BAO data.

B.1 Savage–Dickey density ratio

SSEE is nested within CPL: the SSEE dynamic sector is CPL evaluated at the algebraic point $(w_0, w_a) = (-0.840, -0.670)$. For nested models, the marginal Bayes factor for the null hypothesis $\mathcal{H}_0 : (w_0, w_a) = (-0.840, -0.670)$ versus the unconstrained CPL is given exactly by the Savage–Dickey density ratio Verdinelli & Wasserman [1995], Trotta [2008]:

$$B_{\mathrm{SSEE/CPL}} = \frac{p(w_0^{\mathrm{S}}, w_a^{\mathrm{S}} \mid \text{data, CPL})}{\pi(w_0^{\mathrm{S}}, w_a^{\mathrm{S}} \mid \text{CPL})} = 18.34, \qquad \ln B = +2.91. \tag{18}$$

The CPL posterior is evaluated at $(w_0^{\mathrm{S}}, w_a^{\mathrm{S}})$ using a Scott-bandwidth Gaussian KDE of $N = 500\,000$ thinned CPL chain samples (columns 3–4 of the 5-dimensional CPL chain, shape 2.5×10^6). The marginal prior on (w_0, w_a) is the truncated Gaussian $\mathcal{N}(-1, 0.5^2) \times \mathcal{N}(0, 1^2)$ clipped to the MCMC sampling box $w_0 \in (-2.5, 0.5)$, $w_a \in (-3, 2)$; truncation factors are $(0.9973, 0.9759)$, negligible corrections.

On the Jeffreys scale Trotta [2008] $\ln B = +2.91$ (> 2.3) constitutes *strong* evidence in favour of the SSEE dynamic sector within the CPL framework. The CPL posterior median ($w_0^{\mathrm{CPL}} = -0.826 \pm 0.080$, $w_a^{\mathrm{CPL}} = -0.557 \pm 0.314$) places the SSEE algebraic point at $(0.18\sigma, 0.36\sigma)$, consistent with the 0.24σ joint tension quoted in Section 6.2.

B.2 Ly-α predictive audit ($z = 2.33$)

The Ly-α point at $z = 2.33$ provides the sharpest direct test of SSEE's background expansion. At $z \sim 2$ radiation is negligible and the ratio $D_H/r_d = c\,/\,(H_0\,E(z)\,r_d)$ directly encodes the product $H_0\,E(z)\,r_d$. We evaluate this quantity under two SSEE scenarios and compare to the DESI DR2 measurement $D_H/r_d = 8.632 \pm 0.101$ (Table 1).

Scenario A — raw ssee background ($\Omega_{m,\mathrm{dyn}} = 0.160$, $r_d = 175.6\,\mathrm{Mpc}$). With the algebraic $H_0 = 67.96\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ and $\Omega_m = 0.160$, the CPL dark-energy fraction at $z = 2.33$ is $f_{\mathrm{DE}}(2.33) = 0.648$, giving $E(2.33) = 2.540$ and $D_H(2.33) = 1737\,\mathrm{Mpc}$. The predicted ratio is:

$$\left.\frac{D_H}{r_d}\right|_{z=2.33}^{\mathrm{SSEE,\,raw}} = \frac{1737}{175.6} = 9.89 \quad (12.5\sigma \text{ above DESI}), \tag{19}$$

demonstrating the category error directly: using the cold sector $\Omega_m = 0.160$ in $E(z)$ reduces matter domination at high z, lowers $H(z)$, and enlarges D_H to a 12.5σ discrepancy. This is the same misplacement that produced the spurious background tensions in superseded drafts; the correct total density (Scenario B) resolves it.

Scenario B — ω_m-direct CMB-sector background ($\Omega_{m,\mathrm{CMB}} = 0.30889$, $r_d = 147.17$ Mpc). At $z = 2.33$ the ϕ-DM (mass 40.70 eV, non-relativistic since $z \sim 4.5 \times 10^5$) contributes fully to the background matter density, so the expansion $E(z)$ is set by the CMB-sector value $\Omega_{m,\mathrm{CMB}} = \omega_m/h^2 = 0.30889$ (free-streaming suppresses small-scale clustering, not the homogeneous density). Taking the sound horizon $r_d = 147.17$ Mpc directly from ω_m (CAMB; no $|w_0|$ mapping) gives $E(2.33) = 3.443$, $D_H(2.33) = 1281$ Mpc, and:

$$\frac{D_H}{r_d}\bigg|_{z=2.33}^{\omega_m\text{-direct}} = \frac{1281}{147.17} = 8.71 \quad (0.7\sigma \text{ from DESI}). \tag{20}$$

This supersedes the earlier MIRA-mapped value ($\Omega_m = 0.320$, $D_H/r_d = 8.56$): the matter factor is dissolved (OP-8). The D_M/r_d counterpart at $z = 2.33$ is recovered within 0.1σ. The full 13-point diagonal fit gives $\chi^2 = 12.0$, matching ΛCDM ($\chi^2 = 12.0$): the two backgrounds describe DESI DR2 equally well. The largest SSEE residual is at $z = 0.51$ (D_M/r_d pull -2.1σ); ΛCDM shows a comparable largest residual ($+1.7\sigma$ at $z = 0.706$, D_H/r_d) with the same total χ^2, confirming the residual pattern is a DESI DR2 feature rather than an SSEE-specific failure.

Implication. Eqs. (19)–(20) show that the high-z expansion requires the full CMB-sector matter density $\Omega_{m,\mathrm{CMB}} = 0.30889$, where ϕ-DM counts as non-relativistic matter, rather than the dynamical 0.160: retaining $\Omega_m = 0.160$ in $E(z)$ with $r_d = 147.17$ would predict $D_H/r_d = 11.8$ (a 31σ discrepancy). This is the same two-sector structure that sets the low-z sound horizon (§2.4), now with *no* matter factor (OP-8 dissolved). The Savage–Dickey ratio in B.1 ($\ln B = +2.91$) then holds as a test of the *dynamical sector* under this ω_m-direct background.

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